nature portfolio

Peer Review File

Understanding epistatic networks in the B1 β-lactamases through coevolutionary statistical modeling and deep mutational scanning



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REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

This study of higher ordered epistasis is well-executed and rigorous. In general, the results supported the author's conclusion. I found the paper interesting but was left wondering what was learned that wasn't generally expected. I also struggled to understand why the author's chose homologs that were so distantly related, which I think makes it more difficult to gain insight into epistatic mechanisms. Perhaps it is because they are comparing DMS to DCA (?)

Here are my comments and suggestions for improving the manuscript:

1. Line 23. I expect that entrenched residues would not be easily compensated by changes in nearby interactions. Thus, I recommend deleting the word "However" and instead just start the sentence with "Entrenched."
2. Line 30. The sentence structure makes it sound like it is the organisms that often share <25% identity when what I assume is meant is that the proteins often share <25% identity.
3. Line 33 How do networks of intramolecular interactions enable fitness landscape exploration? Mutations enable exploration. Also, although I share the authors' view that epistasis is important for understanding mutational effects, I do not understand why it is useful to invoke epistasis to explain why deleterious mutations in one protein can be observed in homologs, as the sequences are so different. More on this later.
4. Line 37 "Why can function-destroying…" This general question seems to have been already answered in the previous paragraph: i.e. proteins are complex, and homologs have different sequences. It is certainly not an unanswered question. However, the molecular details of why a particular mutation has that behavior is usually not known and can be difficult to determine.
5. Line 87 The mutational tolerance is a computationally predicted value and may not reflect the actual tolerance. I suggest changing to "predicted mutational tolerance." Also do the same in the legend of Fig 1b and in other places in the manuscript.
6. Line 90 and 91. Is this meaningful? Clades are determined by sequence similarity, and sequence determines the per position prediction of tolerance. Thus, it seems necessarily true different clades will have different patterns. What have we learned?
7. Fig 2E. It might also be good, probably in the supplementary, to compare DMS mutational tolerance to Shannon entropy calculated using the frequency of observations of the 20 amino acids at that position among all the sequences (information theoretical entropy). This would show how much the DCA calculation of Shannon entropy is improving upon just a consideration of the frequencies of amino acids observed at each position (i.e. which has no consideration of co-evolving mutations).
8. Section on epistasis starting at line 178: The two homologs differ in sequence at 70% of the about 220 positions in the protein. That means one sequence must have about 154 mutations to be converted to the other protein. I understand that technically, when a mutation in one protein has a different effect in the other protein that this is epistasis. But it is epistasis involving 154 differences.

Why are these proteins useful for studying epistasis? What is learned by studying such a high-ordered epistasis that can be better or more easily studies using much lower ordered epistasis (pairwise, tertiary, and a few steps above that)? Also, I think it would be useful to show how magnitude of fitness effect and epistasis values are related.
9. Line 187 I don't fine it "striking" that there are so many different mutational effects between VIM-2 and NDM-1 when the proteins have 70% different sequences. I find it expected. If the intent was just to emphasize how different they are, perhaps just say "very different" or "quite different."
10. Line 187 is this 28% of all residues or 28% of residues that are classified as "shared?" It might be good to enumerate how many residues got classified as shared, delineating how much because both proteins had the same amino acid at that position and how much because the mutation of the other's WT residue was neutral (is this what is meant by "'compatible', 50 positions" on line 186)
11. Last section of results. The authors study a system of epistasis involving about 154 different amino acids. In this section, as an example, they look at pairwise and tertiary epistasis. I understand the reasons (too complex to look at epistasis involving potentially that many positions). But this begs the question of whether studying epistasis in such a complex system is the best choice. Wouldn't it have been better to do an extensive study of pairwise, tertiary, etc. epistasis? I am not clear on the reasons the authors chose their system (such distant homologs) and what was learned that could not have been learned in a simpler, more tractable study of lower-order epistasis. It would be helpful if the authors explained this (or explain it more clearly if I have missed it).
12. Line 388. Should be 10^-5 not 10^5

Reviewer #2 (Remarks to the Author):

The authors perform an in-depth analysis of mutational landscapes of the B1 metallo-beta-lactamases enzyme family, combining a co-evolutionary statistical approach (DCA) to deep mutational scanning (DMS) experiments. The work contains a substantial amount of new results, all presented in carefully designed plots, which deserve publication for a biological interdisciplinary audience. I have only a few remarks of presentation and questions.

Lines 59-61: I think the motivation for combining DCA and DMS could be strengthened, for example the authors could consider if it is worth stressing that the methods allow one to adopt a double perspective, one statistical hence more global across homologs and evolutionary pressures (but relying on a computational inference step), the other one probing directly mutational effects but specific to two homologs.

Fig 1b: the authors could consider indicating in the figure that the bar with arrows corresponds to VIM-2

Lines 70-71: why is this family an excellent example for epistatic networks? Could the authors add a comment and/or references?

Line 82: the 100 homologs are chosen to minimise sequence identity, could the authors add in the methods how they made this choice more precisely?

Line 84: the authors introduce the CDE, but in the previous references it seems to me that it was not based on probabilities relative to the WT. Could the authors clarify this connection in the methods?

Line 155: the authors checked the correlation to ASA. Did the authors check the correlation to other residue properties? Even if they found no correlation, could it be worth to discuss what properties they investigated and the fact that none of them accounts for mutational effects as much as ASA?

Figure 4a: here there are two terms that do not seem fully explained in main and methods. First 'wt reversion', which is also recalled in the methods without explanation: does it come from the swapping of residues between wild-types at differing positions? If yes, it needs to be stated clearly. The second term is 'calculate epistatic': while epistasis per se is explained, I would have expected in the methods to see the formulas for this calculation in DCA and DMS but I do not seem to see them. This definitely needs to be included.

Figure 4b: could the authors clarify in the methods that their statistical significance analysis lead to the 1.96 x SD criterion? Then I would refer to the methods explicitly here. Also the acronym SD wasn't defined I think.

Figure 4d: how was the fraction of epistatic calculated?

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Line 243: refer already here to the figure supporting the statement

Line 263-264: could one add concrete examples of 'peculiarities'

Methods:

In the methods, I struggled a bit to understand what methodological step refers to what in the main. I would suggest that each section of the methods starts with a sentence introducing for what step of the analysis the approach has been used. For example, regarding the first two sections, are they meant to explain how the datasets were assembled respectively for the training and evaluation of DCA?

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Some formulas lack a bit of explanation.
Eq 1: one needs to clarify what each term stands for.

Clarify what P(pos,mut) stands for in (2), given that this was called P(mutant) in the main.

What is n in (3)?

In (6), one should have mut=1 in the sum

In lines 426-28, where the terms 'epistatic' and 'non-epistatic' exchanged? If not, could you explain why?

Reviewer #3 (Remarks to the Author):

The authors make creative use of direct coupling analysis (DCA) to probe epistatic drift - changes in the effects of mutations due to epistatic interactions with substitutions that accrue during evolution. Experimental characterization of epistatic drift requires applying deep mutational scanning (DMS) on multiple homologs, and has therefore been limited to at most a dozen homologs in a couple of protein families. By computationally predicting the effects of mutations using DCA, the authors analyze epistatic drift across a hundred sequences. In line with previous studies, they find that epistatic drift is widespread.

While the use of DCA to study epistatic drift is novel, I have significant concerns about the analyses underlying the claims of widespread epistatic drift.

1. The authors' most significant finding is that mutational tolerance at the same residue varies substantially among homologs, which they interpret as evidence of widespread epistatic drift. However, mutational tolerance can vary even in the absence of epistasis. Suppose leucine is better fit than alanine at a site. Homologs with alanine at the site will have higher mutational tolerance than homologs with leucine, simply because they have a suboptimal wild-type state. Variation in mutational tolerance is expected whenever the wild-type amino acids vary in fitness. The authors have not controlled for this variation in all of their analyses in Figs. 1-3. They do control for this variation in Fig. 4, where they compare the effects of the *same* mutations on two genetic

backgrounds and find pervasive epistasis. This control should be extended to Figs. 1-3. Before then, the changes in mutational tolerance they find cannot be linked to epistatic drift.

2. Another concern relates to the accuracy of DCA. The authors show that the mutational tolerance predicted by DCA and measured by DMS have a rather weak correlation of 0.55 (Fig. 2e). Given this, how accurate is the inference of mutational tolerance in Fig. 1? When the inferred mutational tolerance differs between two homologs, how can we tell whether the difference is noise or real?

3. A recurring claim is that epistatic drift is most pronounced at residues of intermediate burial, but the data are not presented to clearly demonstrate this pattern. In Fig. 3a-c, the extent of epistatic drift at each residue is shown as the thickness in protein structure, but the thick regions appear spread across the structure with no obvious relation to buriedness. In Fig. 3d-f, the extent of epistatic drift is shown on a color scale, so it is difficult to discern whether any trend is statistically meaningful. To justify their claim, the authors should plot the residue buriedness on the x-axis and the extent of epistatic drift on the y-axis and perform statistical testing. Such a plot is shown in Fig. 4e for the analysis of DMS data but without statistical testing.

4. That epistasis exists or is widespread is no longer a novel finding because dozens of studies have illustrated it across numerous protein families. While the use of DCA to study epistatic drift is novel (pending evaluation of its quantitative accuracy), the principal finding of the study is that epistatic drift is widespread. The authors should be clear and explicit on how their findings add to the existing literature and why researchers should be interested in them when the prevalence of epistasis has been so firmly established.

Minor comments

The abstract reads as if DCA was performed on just 100 sequences (line 17), which is too few for reliable analysis, when in reality it was performed on thousands of sequences.

This important claim has no analysis to back it up: "homologs closer in the phylogeny show similarity in mutation patterns, and different clades present different patterns" (lines 90-91). This is not obvious from a visual inspection of Fig. 1b, so the authors should provide an explicit analysis.

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Please see the above replies.

1. Line 23. I expect that entrenched residues would not be easily compensated by changes in nearby interactions. Thus, I recommend deleting the word "However" and instead just start the sentence with "Entrenched."

Abstract updated as suggested.

2. Line 30. The sentence structure makes it sound like it is the organisms that often share <25% identity when what I assume is meant is that the proteins often share <25% identity.

Rephrased to emphasize the % identity refers to the homologs.

3. Line 33 How do networks of intramolecular interactions enable fitness landscape exploration? Mutations enable exploration. Also, although I share the authors' view that epistasis is important for understanding mutational effects, I do not understand why it is useful to invoke epistasis to explain why deleterious mutations in one protein can be observed in homologs, as the sequences are so different. More on this later.

We have tried to clarify this in the intro. While mutations are what confer movement through sequence space, only viable paths can be traversed. Epistasis is key to sequence space exploration as it is what allows otherwise deleterious mutations (in the context of the WT) to be sampled. Hence, we believe the fixation of deleterious mutations to be a hallmark of epistasis, *i.e.*, the same mutation has different effects in different backgrounds (either due to specific or non-specific interactions). We understand there to be a large number of mutational differences between backgrounds, but these differences are in essence the compounding of epistatic intramolecular interactions.

4. Line 37 "Why can function-destroying…" This general question seems to have been already answered in the previous paragraph: i.e. proteins are complex, and homologs have different sequences. It is certainly not an unanswered question. However, the

molecular details of why a particular mutation has that behavior is usually not known and can be difficult to determine.

We have changed 'why' to 'what causes' to better reflect the fact that they are already fixed in nature, but we do not know the particular details.

5. Line 87 The mutational tolerance is a computationally predicted value and may not reflect the actual tolerance. I suggest changing to "predicted mutational tolerance." Also do the same in the legend of Fig 1b and in other places in the manuscript.

We have updated the legend and the main text introducing the calculations to emphasize the DCA mutational tolerance is predicted. However, as the term is used very frequently, we refrain from prefixing all instances with 'predicted' and instead just emphasize when the term is introduced. Later on, the predicted mutational tolerance is naturally disambiguated by the DCA prefix.

6. Line 90 and 91. Is this meaningful? Clades are determined by sequence similarity, and sequence determines the per position prediction of tolerance. Thus, it seems necessarily true different clades will have different patterns. What have we learned?

We have modified the section in the main text, and have added a supporting supplementary figure (now **Supp. Fig. 1**). We wish to emphasize that there is a difference between expecting different clades to have different patterns, and predicting which pattern is present in which clade. Our approach can quantify the different tolerance to mutations between sequences and positions. Hence, we have explicitly illustrated the ability to detect mutational differences between sequences (**Supp. Fig. 1b**) across all 100 homologs in terms of sequence divergence, as well as to predict position-specific differences (**Supp. Fig. 1c**) in mutational tolerance. The changes to the main text also aim to emphasize the goal of and ability to detect specific differences.

7. Fig 2E. It might also be good, probably in the supplementary, to compare DMS mutational tolerance to Shannon entropy calculated using the frequency of observations of the 20 amino acids at that position among all the sequences (information theoretical entropy). This would show how much the DCA calculation of Shannon entropy is improving upon just a consideration of the frequencies of amino acids observed at each position (i.e. which has no consideration of co-evolving mutations).

We have added comparison of the information theoretical entropy (labelled context independent entropy, or CIE) to the main text (page 8) and supplementary figures (now **Supp. Fig. 4**). Indeed, consideration of sequence context provides better predictions of the observed mutational tolerance than using the positional frequencies alone (ΔR^2 = 0.09 for NDM-1 and ΔR^2 = 0.07 for VIM-2). We also emphasize that the analysis of specific homolog behaviors would not be possible using CIE, as it ignores sequence context by its nature and generates a single value per site regardless of sequence background.

8. Section on epistasis starting at line 178: The two homologs differ in sequence at 70% of the about 220 positions in the protein. That means one sequence must have about 154 mutations to be converted to the other protein. I understand that technically, when a mutation in one protein has a different effect in the other protein that this is epistasis. But it is epistasis involving 154 differences. Why are these proteins useful for studying epistasis? What is learned by studying such a high-ordered epistasis that can be better or more easily studies using much lower ordered epistasis (pairwise, tertiary, and a few steps above that)? Also, I think it would be useful to show how magnitude of fitness effect and epistasis values are related.

The utility of such a system is described by the reply at the beginning. As an added note, we believe there is a general confusion between collective effects and higher-order epistasis in this comment. The fact that we have 132 mutations between WT backgrounds does NOT imply that we are looking to 132nd order epistasis. We just declare that epistasis is collective and even if described as pairwise, only the cumulation of many pairwise weak but coherent distributions leads to strong context dependence.

The magnitude of fitness effects and the epistasis can be observed from Fig. 4b, where the fitness of a mutation in each background can be referenced with its distance from the diagonal (representing no epistasis).

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Rephrased to 'very different'.

10. Line 187 is this 28% of all residues or 28% of residues that are classified as "shared?" It might be good to enumerate how many residues got classified as shared, delineating how much because both proteins had the same amino acid at that position and how much because the mutation of the other's WT residue was neutral (is this what is meant by "'compatible', 50 positions" on line 186)

The 28% was clarified as the individual mutations being compared across the shared positions. Additional clarification of the number of shared positions has been added, including the total (131) as well as the breakdown from having the same WT (81) or being 'compatible' (50).

11. Last section of results. The authors study a system of epistasis involving about 154 different amino acids. In this section, as an example, they look at pairwise and tertiary epistasis. I understand the reasons (too complex to look at epistasis involving potentially that many positions). But this begs the question of whether studying epistasis in such a complex system is the best choice. Wouldn't it have been better to do an extensive study

of pairwise, tertiary, etc. epistasis? I am not clear on the reasons the authors chose their system (such distant homologs) and what was learned that could not have been learned in a simpler, more tractable study of lower-order epistasis. It would be helpful if the authors explained this (or explain it more clearly if I have missed it).

Please see the reply at the beginning. In summary, we have chosen such a system specifically because it encapsulates the complexities that arise from highly diverged sequences. We also believe that studies of lower-order epistasis that cover a complete set of combinations but remain sparse in terms of sequence coverage, would not extrapolate well to the collective effects of so many mutations. Hence, we focus on methods to study the final collective effect of the compounded mutations.

12. Line 388. Should be 10^-5 not 10^5

Corrected.

Reviewer #2 (Remarks to the Author):

The authors perform an in-depth analysis of mutational landscapes of the B1 metallo-beta-lactamases enzyme family, combining a co-evolutionary statistical approach (DCA) to deep mutational scanning (DMS) experiments. The work contains a substantial amount of new results, all presented in carefully designed plots, which deserve publication for a biological interdisciplinary audience. I have only a few remarks of presentation and questions.

Lines 59-61: I think the motivation for combining DCA and DMS could be strengthened, for example the authors could consider if it is worth stressing that the methods allow one to adopt a double perspective, one statistical hence more global across homologs and evolutionary pressures (but relying on a computational inference step), the other one probing directly mutational effects but specific to two homologs.

Indeed, such are the strengths attributable to each method. We have now addressed the motivation in more detail in the introduction.

Fig 1b: the authors could consider indicating in the figure that the bar with arrows corresponds to VIM-2

Additional labels have been added to clarify the VIM-2 secondary structure.

Lines 70-71: why is this family an excellent example for epistatic networks? Could the authors add a comment and/or references?

Related to the reply at the start, this family is excellent because of its highly diverged nature, and we have tried to clarify this in text (page 3 intro. and page 5). Of course, it does not necessarily have to be this family (any diversified protein family with assayable function will work), but the fact that it has a good selectable function and functionally characterized diverged homologs as DMS starting points makes it a suitable choice.

Line 82: the 100 homologs are chosen to minimise sequence identity, could the authors add in the methods how they made this choice more precisely?

We have clarified this in a number of ways. We now state the diversity of the sequences directly (~70% sequence divergence) and show a plot of the distribution of pairwise divergence between homologs in **Supp. Fig. 1a**. The method for selecting homologs has also been added to the methods section on DCA.

Line 84: the authors introduce the CDE, but in the previous references it seems to me that it was not based on probabilities relative to the WT. Could the authors clarify this connection in the methods?

CDE was introduced in Vigué et al. using exactly the same definition, using a reference sequence which in that case was one specific *E. coli* strain.

Line 155: the authors checked the correlation to ASA. Did the authors check the correlation to other residue properties? Even if they found no correlation, could it be worth to discuss what properties they investigated and the fact that none of them accounts for mutational effects as much as ASA?

We did not check other residue properties in this case, as we had previously investigated such properties in detail in correlation to the mutational effects in VIM-2 by linear modeling (PMID: 32510322). Of the various properties, those related to structure gave the strongest explanation in fitness score variance (ASA, ΔΔG), while residue polarity or size generally explained less than 10% of the variance. We have added an explanation in the main text at the beginning of this section.

Figure 4a: here there are two terms that do not seem fully explained in main and methods. First 'wt reversion', which is also recalled in the methods without explanation: does it come from the swapping of residues between wild-types at differing positions? If yes, it needs to be stated clearly. The second term is 'calculate epistatic': while epistasis per se is explained, I would have expected in the methods to see the formulas for this calculation in DCA and DMS but I do not seem to see them. This definitely needs to be included.

We have tried to clarify this in the main text. Indeed, 'wt reversion' refers to the swapping of WT residues at homologous positions with differing WT amino acids between NDM-1 and VIM-2. For 'calculate epistatic', it has been updated to 'test for epistasis' and the main text has

been rearranged to better emphasize this was done by a z-testing for significant differences between a mutation's effect in the NDM-1 and VIM-2 backgrounds against a null distribution with the same standard deviation as the NDM-1 synonymous variant distribution. These tests were only applied for the DMS data set. Both are also described in the methods, but as the z-test is a standard and well known test, we did not include an equation for it.

Figure 4b: could the authors clarify in the methods that their statistical significance analysis lead to the 1.96 x SD criterion? Then I would refer to the methods explicitly here. Also the acronym SD wasn't defined I think.

SD has been expanded to 'standard deviation'. We wish to clarify that the 1.96xSD criterion outlined in the figures are not tied to the statistical significance tests. We conduct the statistical tests only on the differences between NDM-1 and VIM-2 to determine if the mutants in shared positions are epistatic; the tests relied on FDR correction instead of a strict range of the standard deviation. Here, the lines are intended as a visual guide to the regions of neutral effects that would be expected of each homolog (axes) as well as when there is no epistasis (diagonal). We believe the decision to use the 1.96xSD is a fairly standard way to represent the range that would capture ~95% of a null or baseline distribution.

Figure 4d: how was the fraction of epistatic calculated?

Fraction of epistatic mutations per position is calculated as $N_{\text{mut epistatic}}/N_{\text{total observed}}$. We have added this to the main text and figure legend.

Figure 4f: could the authors use the same terminology in caption and figure? It seems that wt comes after or before the protein name a bit too interchangeably.

The labels for Fig. 4f and 4h have been made more explicit, and the legend has been adjusted.

Figure 4g: can one have a colorbar?

Added color legend.

Figure 4h: I think here the authors might consider to indicate in the figure itself that above/below diagonal indicates gain/loss of fitness.

We do not make this distinction as the diagonal does not indicate the separation between gain or loss of fitness when swapping WT residues between backgrounds. The gain or loss of "fitness" (ΔE is not a direct equivalent of fitness) of swapping a WT residue into either of the two backgrounds would be indicated on the x- or y- axis. Points above the diagonal would be cases where WT residues tend to be mutually compatible across the 2 backgrounds, and points below the diagonal are where WT residues tend to be mutually incompatible (*i.e.*, entrenched).

Line 215: can one refer again to the figures supporting this?

Reference to Fig. 4i added.

Figure 5d: I would not use CDE in the caption here, since in the figure the same quantity is referred to as mutational tolerance

Updated ‘CDE’ to ‘mutational tolerance’.

Figure 5f: can one mark the triple mutant?

The triple mutant is now marked in purple.

Line 243: refer already here to the figure supporting the statement

Reference to Fig. 5e added.

Line 263-264: could one add concrete examples of ‘peculiarities’

Clarified the ‘peculiarities’ to be differences between the global model and more background specific behaviors, with reference to Fig. 2e.

Methods:

In the methods, I struggled a bit to understand what methodological step refers to what in the main. I would suggest that each section of the methods starts with a sentence introducing for what step of the analysis the approach has been used. For example, regarding the first two sections, are they meant to explain how the datasets were assembled respectively for the training and evaluation of DCA?

As suggested, we have added leading statements for each method section to clarify their purpose.

Line 307: how was the mis-alignment evaluated in the manual check?

The manual check was to exclude poor alignments that were obvious by eye, including alignments where most of the sequence is shifted or out-of-place relative to most sequences or where large gaps exist in generally conserved columns. This has been added to the methods.

Some formulas lack a bit of explanation.
Eq 1: one needs to clarify what each term stands for.

Added additional clarification on the terms in equation (1).

Clarify what P(pos,mut) stands for in (2), given that this was called P(mutant) in the main.

We have changed the P(mutant) to P(mut, pos) for consistency, and have clarified that this is the probability of a mutant at a given position in the methods.

What is n in (3)?

Updated ‘n’ to ‘n=20’, which is all amino acids (including WT).

In (6), one should have mut=1 in the sum

Updated to ‘mut=1’.

In lines 426-28, where the terms ‘epistatic’ and ‘non-epistatic’ exchanged? If not, could you explain why?

The terms were not exchanged, though we agree it can be confusing. In a reversion pairing where reversion in one background is negative, there would be epistasis if the reversion in the other background is neutral. However, if the other reversion is positive (and of similar magnitude) this matches the expectations for non-epistasis. We have tried to clarify this in the methods.

Reviewer #3 (Remarks to the Author):

The authors make creative use of direct coupling analysis (DCA) to probe epistatic drift - changes in the effects of mutations due to epistatic interactions with substitutions that accrue during evolution. Experimental characterization of epistatic drift requires applying deep mutational scanning (DMS) on multiple homologs, and has therefore been limited to at most a dozen homologs in a couple of protein families. By computationally predicting the effects of mutations using DCA, the authors analyze epistatic drift across a hundred sequences. In line with previous studies, they find that epistatic drift is widespread.

While the use of DCA to study epistatic drift is novel, I have significant concerns about the analyses underlying the claims of widespread epistatic drift.

1. The authors’ most significant finding is that mutational tolerance at the same residue varies substantially among homologs, which they interpret as evidence of widespread epistatic drift. However, mutational tolerance can vary even in the absence of epistasis. Suppose leucine is better fit than alanine at a site. Homologs with alanine at the site will have higher mutational tolerance than homologs with leucine, simply because they have a suboptimal wild-type state. Variation in mutational tolerance is expected whenever the wild-type amino acids vary in fitness. The authors have not controlled for this variation in all of their analyses in Figs. 1-3. They do control for this variation in Fig. 4, where they

compare the effects of the *same* mutations on two genetic backgrounds and find pervasive epistasis. This control should be extended to Figs. 1-3. Before then, the changes in mutational tolerance they find cannot be linked to epistatic drift.

If we understand correctly, this comment is stating that more mutations will have higher fitness if the starting amino acid is less fit, and hence mutating to other amino acids will appear to be more beneficial (hence appear more mutationally tolerant). However, as the mutational tolerance calculations are applied to all amino acids at a position (including WT), the variation in the starting WT amino acid is already accounted for. The above would only be an issue in a hypothetical formulation of mutational tolerance that is calculated using individual differences to the WT fitness, but the Shannon Entropy calculations normalize all possible amino acid states as probabilities (relative to all amino acids, not just WT). For example, if two sets of 20 amino acids have the same probabilities, but have different designations of the 'WT' amino acid, both would still give the same mutational tolerance when calculated. Hence, the differences we see are due to the epistasis of different genetic contexts. We have added a clarifying statement on page 5.

We note the extra control for matching the same mutations in Fig. 4 is because we are then analyzing individual mutational differences, rather than positional mutational tolerance. In this case, a direct comparison of relative differences necessitates the extra control. This has also been clarified in text on page 13.

2. Another concern relates to the accuracy of DCA. The authors show that the mutational tolerance predicted by DCA and measured by DMS have a rather weak correlation of 0.55 (Fig. 2e). Given this, how accurate is the inference of mutational tolerance in Fig. 1? When the inferred mutational tolerance differs between two homologs, how can we tell whether the difference is noise or real?

The ability of DCA to systematically capture the effect of epistasis on mutations in proteins has been thoroughly demonstrated in a variety of applications. This includes the ability to predict DMS data (PMID: 26446903), E. coli strain polymorphisms (PMID: 29073099, 35821377), modeling of evolution experiments (PMID: 34751386) and generating highly diverged but function artificial sequences up to 70-80% divergence (PMID: 32703877). Hence, even if noisy in nature, we build on such previous observations the ability of DCA to meaningfully capture epistasis in protein sequences.

Although the correlation of DCA and DMS mutational tolerance is only 0.55, we do not believe noise to be the primary contributor. As discussed in the text, the mutational tolerances captured by DCA and DMS are likely of different natures. DCA is a global model that captures aggregated selective pressures across the evolution of the family based on sequence coevolution, while DMS in this system is an experiment that measures survival conferred to a host by variants of specific homologs with further specifics involving the type of antibiotic and level of selection pressure. Even with a "perfect" DCA model, we do not expect there to be perfect correlation as the experiment and model capture fundamentally different facets of information.

Finally, even if there may be noise in the DCA predictions, we believe our use of the method mitigates the effects of noise on our findings as we largely use DCA to examine broad level trends rather than exact differences between specific data points. Despite the potential noise in single predictions, we have shown that trends from both DCA and DMS tend to agree quite well (*e.g.*, Fig. 3), even when directly considering just differences between NDM-1 and VIM-2 (Fig. 3c and f, Fig. 4h and i).

3. A recurring claim is that epistatic drift is most pronounced at residues of intermediate burial, but the data are not presented to clearly demonstrate this pattern. In Fig. 3a-c, the extent of epistatic drift at each residue is shown as the thickness in protein structure, but the thick regions appear spread across the structure with no obvious relation to buriedness. In Fig. 3d-f, the extent of epistatic drift is shown on a color scale, so it is difficult to discern whether any trend is statistically meaningful. To justify their claim, the authors should plot the residue buriedness on the x-axis and the extent of epistatic drift on the y-axis and perform statistical testing. Such a plot is shown in Fig. 4e for the analysis of DMS data but without statistical testing.

As suggested, we have produced plots similar to Fig. 4e for the data in Fig. 3d-f, now in Supplementary Fig. 3. In addition, we also plot the data in each ASA class (buried, intermediate, exposed) directly as distributions, and we perform a one sided Brunner-Munzel test on all pairs of distributions to determine if any of the distributions tend to have larger values than the other, displayed along with the distributions. Additional comments have been added to the main text on page 11.

The data in Fig. 4e has also been plotted as distributions by ASA class and statistically tested in Supplementary Fig 4a, with an added comment on page 14.

We have also clarified in the intro and discussion that the common finding being that ‘residues with the strongest epistatic effects’ are observed at intermediate burial. This is what we originally intended, but we see how it could have been confused for claiming that the distributions of values in regions of intermediate burial also tend to have higher values (though this is still supported in the case of DCA mutational tolerance and individual mutant epistasis in DMS).

4. That epistasis exists or is widespread is no longer a novel finding because dozens of studies have illustrated it across numerous protein families. While the use of DCA to study epistatic drift is novel (pending evaluation of its quantitative accuracy), the principal finding of the study is that epistatic drift is widespread. The authors should be clear and explicit on how their findings add to the existing literature and why researchers should be interested in them when the prevalence of epistasis has been so firmly established.

Please see the reply at the beginning. Despite widespread epistasis being an observation in our study, it is far from the principal finding of our study. We reveal the specific patterns of epistasis on very distant homologs, provide novel findings with regards to the

predictability of epistasis, and explore specific interactions between entrenched residues. As stated in the reply, we believe these new findings were already explicitly stated in the text.

Minor comments

The abstract reads as if DCA was performed on just 100 sequences (line 17), which is too few for reliable analysis, when in reality it was performed on thousands of sequences.

We have clarified that the 100 sequences were the subject of 'in silico DMS' rather than directly for DCA in the abstract.

This important claim has no analysis to back it up: "homologs closer in the phylogeny show similarity in mutation patterns, and different clades present different patterns" (lines 90-91). This is not obvious from a visual inspection of Fig. 1b, so the authors should provide an explicit analysis.

We have modified this section to better communicate our intent, which is to show that the measures of mutational tolerance can reveal specific patterns across the homologs. We have added a supplementary figure (**Supp. Fig. 1b**) that explicitly quantifies similarity in mutational tolerance with regard to sequence divergence, as suggested.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The authors have satisfactorily address my comments.

Reviewer #2 (Remarks to the Author):

I am happy with the changes and the answers by the authors.

Reviewer #3 (Remarks to the Author):

1. Regarding our previous comment 1, the authors are aware of our concern and have explained that their analysis is not affected by it. We believe in the authors' explanation, but there remains a discrepancy between their explanation and their method as written on p. 21. Mutational tolerance as defined in Eq. (2) and (3) on p. 21 is indeed sensitive to the choice of wild-type amino acid (see our supplemental note). Our guess is that the authors forgot to mention a normalization step that would make the probabilities (and the mutational tolerance) indeed independent of the choice of wild-type amino acid.

Taking the authors' explanation and our guess as true, however, "mutational tolerance" is a misleading term. With DCA, the authors are not analyzing any mutations but rather the sequence constraint at each site in each homolog. Using "sequence constraint" in place of "mutational tolerance" would have prevented us from raising this comment at all.

2. We appreciate the authors' careful discussion of the powers and limits of DCA, which answers our comment 2. We believe some of the response should be presented in the main text. Discussing the limits of DCA and how the study's main conclusions may be robust to them is important because this is the first study to use DCA for the purpose.

3. We appreciate the additional analyses that the authors have performed in response to our comment 3.

4. We have significant concerns about how the authors have responded to the questions of novelty. Two reviewers have questioned what novel findings this paper have produced beyond the well-established fact of widespread epistasis, which should be taken as indication that the novelties of the work have not been clearly delivered. It is therefore frustrating that the authors believe they

“were already explicitly stated in the text” and decline to make any changes to address this point.

Besides, the authors have summarized five novelties in their rebuttal (point 2), but we see only one as genuinely novel.

1) “we show that the effect of some of those mutations are predictable (Fig. 2)”

Fig. 2 does not predict the effect of any mutation. Fig. 2e compares experimentally measured and DCA-predicted mutational tolerance values, which show a correlation of 0.55. There is nothing in the abstract or introduction that points to the ability to predict mutational effects as a novelty (or even a subject) of the paper.

2) “propose a general pattern involving the structural position of epistatic residues”

The authors’ finding is that strongest changes in mutational tolerance are observed preferentially in residues with intermediate burial. This is novel and has a plausible structural explanation.

3) “identify specific entrenched residues between the two homologs”

Identification of entrenched residues between homologous proteins is not novel (see PMID 29626131 and 20975933 for example). It is new for the two particular beta-lactamases, but our question is what can be generalized beyond the two proteins. Furthermore, the authors haven’t really shown entrenchment: entrenchment occurs when an initially reversible substitution becomes irreversible because the reversion develops an epistatic incompatibility with intervening substitutions. Diagnosing entrenchment therefore requires phylogenetic ordering of substitutions in time, not an unordered comparison between two extant sequences.

4) “compensations of entrenched residues are not trivially pairwise … (Fig. 5e)”

Recent papers make it clear that mutations interact with many epistatic partners. (The authors cite such papers in the description of Fig. 5.) This means that a long-term change in the effect of a mutation is typically attributable to many substitutions, not a single or even a few. The authors’ finding is fully in line with this knowledge.

5) “the extent of heterogeneity in mutational effects uncovered by DCA and backed up by DMS is unprecedented”

We do not fully understand what “unprecedented” means, but here are two studies that show pervasive epistasis among distantly related homologs. PMID 29626131 compares two proteins that differ at 168 out of 365 sites (46% divergent), and finds that 63 of the 168 differences cause incompatibility. PMID 20975933 compares homologs that are up to 50% divergent and finds that the effects of a large set of mutations have become randomized. In terms of the extent of epistasis, it is not clear to us what is “unprecedented” about this study.

Overall, it is not that we doubt the value of this work. Far from it, we think the deployment of DCA and its conjunction with DMS is an exciting methodological direction. But it is the authors' responsibility to fully develop the implications of their findings. In our opinion, the field of epistasis is inundated with studies that simply report the existence and prevalence of epistasis. It would be a service to the field for leaders such as the present authors to push beyond the mannerism and show what conceptual advances can be made.

Let $P(\alpha)$ denote the DCA probability of amino acid α at a site in a sequence, and let WT be the wild-type amino acid. Eq. (2) in p. 21 defines the relative probability of α as

$$P_{rel}(\alpha) = \frac{P(\alpha)}{P(WT)}.$$

This quantity measures the likelihood of α relative to WT and is therefore dependent on the choice of wild-type amino acid. Eq. (3) then defines mutational tolerance as

$$CDE = -\sum_{\alpha} P_{rel}(\alpha) \log_2 P_{rel}(\alpha),$$

which can be rearranged to

$$CDE = -\frac{1}{P(WT)}\left[\sum_{\alpha} P(\alpha) \log_2 P(\alpha) - \log_2 P(WT) \sum_{\alpha} P(\alpha)\right].$$

As far as I can tell, this definition of CDE depends on the choice of WT.

My guess is that the authors forgot to mention that P_{rel} is normalized to sum to 1 across all amino acids. The actual probability used for the computation of CDE is

$$\widetilde{P_{rel}}(\alpha) = \frac{P_{rel}(\alpha)}{\sum_{\alpha} P_{rel}(\alpha)}.$$

Unlike P_{rel}, which does not sum to 1 across amino acids and is therefore not a probability, $\widetilde{P_{rel}}$ is a probability. Moreover, $\widetilde{P_{rel}}$ is independent of the choice of wild-type amino acid because $P(WT)$ is canceled out by the normalization. Is this what the authors have done?

Reviewer #3 (Remarks to the Author):

1. Regarding our previous comment 1, the authors are aware of our concern and have explained that their analysis is not affected by it. We believe in the authors' explanation, but there remains a discrepancy between their explanation and their method as written on p. 21. Mutational tolerance as defined in Eq. (2) and (3) on p. 21 is indeed sensitive to the choice of wild-type amino acid (see our supplemental note). Our guess is that the authors forgot to mention a normalization step that would make the probabilities (and the mutational tolerance) indeed independent of the choice of wild-type amino acid.

Taking the authors' explanation and our guess as true, however, "mutational tolerance" is a misleading term. With DCA, the authors are not analyzing any mutations but rather the sequence constraint at each site in each homolog. Using "sequence constraint" in place of "mutational tolerance" would have prevented us from raising this comment at all.

The guess by the referee is correct, we forgot to mention that the probabilities in Eq.2 need to be normalized before they are inserted in Eq.3. We have now fixed the problem, and we apologize for the mistake that led to this confusion.

2. We appreciate the authors' careful discussion of the powers and limits of DCA, which answers our comment 2. We believe some of the response should be presented in the main text. Discussing the limits of DCA and how the study's main conclusions may be robust to them is important because this is the first study to use DCA for the purpose.

We have incorporated the response into the first section of the discussion.

3. We appreciate the additional analyses that the authors have performed in response to our comment 3.

4. We have significant concerns about how the authors have responded to the questions of novelty. Two reviewers have questioned what novel findings this paper have produced beyond the well-established fact of widespread epistasis, which should be taken as indication that the novelties of the work have not been clearly delivered. It is therefore frustrating that the authors believe they "were already explicitly stated in the text" and decline to make any changes to address this point.

In our first reply, we now see we may have grouped two similar but distinct concerns under the response of the 'expectedness of widespread epistasis'. Reviewer 1, the other reviewer with comments on expectedness of epistasis had framed it commonly as 'epistasis would not be surprising between such distant homologs' rather than strictly a concern about novelty. We have since satisfied reviewer 1's comments with the explanation that we engage with the specifics patterns of such epistasis, which is very different from simply expecting epistasis to be present. As a result, we may have made our list of specific observations appear as a list of novelties unintentionally.

Regardless, we now wish to better address the present reviewer's concerns on novelty. Firstly, we wish to emphasize that the list of findings were not intended as statements on novelty, and we agree with the reviewer's notes below regarding existing

findings. We have tried to better reflect what is existing knowledge in the field, with details below.

In regards to specific novel contributions, we believe the main methodological novelty is in our combining of the use of computational (DCA) and experimental (DMS) methods for studying epistasis, which is discussed in the introduction and discussion. As noted by the reviewer, we have also discovered the tendency for the strongest epistatic effects to be observed in residues of intermediate burial, which is a novel finding. The finding is already discussed numerous times in the results and in the discussion, and we have tried to make this finding more prominent in the abstract in this revision.

We address each specific point below:

Besides, the authors have summarized five novelties in their rebuttal (point 2), but we see only one as genuinely novel.

1) “we show that the effect of some of those mutations are predictable (Fig. 2)”

Fig. 2 does not predict the effect of any mutation. Fig. 2e compares experimentally measured and DCA-predicted mutational tolerance values, which show a correlation of 0.55. There is nothing in the abstract or introduction that points to the ability to predict mutational effects as a novelty (or even a subject) of the paper.

This was poorly phrased. In reference to analyzing specific epistatic patterns, we meant to highlight that a DCA model built on sequence data only can generate predictions on epistasis that correlate with experimental data. Rather than “effect of some … mutations” which sounds like individual mutations, we should have simply stated “mutational tolerance”.

2) “propose a general pattern involving the structural position of epistatic residues”

The authors’ finding is that strongest changes in mutational tolerance are observed preferentially in residues with intermediate burial. This is novel and has a plausible structural explanation.

This is one of our novelties as noted above, we have tried to better highlight this finding in the abstract.

3) “identify specific entrenched residues between the two homologs”

Identification of entrenched residues between homologous proteins is not novel (see PMID 29626131 and 20975933 for example). It is new for the two particular beta-lactamases, but our question is what can be generalized beyond the two proteins.

Indeed the detection of entrenchment is not novel, to better highlight existing work, we’ve added references to previous work on evolutionary entrenchment and contingency in the first paragraph of the introduction (PMIDs 29626131, 26056312 and 34061027; PMID 20975933 was already included in numerous places).

Furthermore, the authors haven’t really shown entrenchment: entrenchment occurs when an initially reversible substitution becomes irreversible because the reversion develops an epistatic incompatibility with intervening substitutions. Diagnosing

entrenchment therefore requires phylogenetic ordering of substitutions in time, not an unordered comparison between two extant sequences.

We agree that identifying specific cases of entrenchment entails determining the order of the mutations. However, we still believe “entrenched residues” to be an appropriate description of what we observe. In our case, the mutational differences have already accrued and any effects on the reversion between existing WT backgrounds will be as a result of mutations that have already taken place. A residue is effectively “entrenched” in one WT by all the accompanying mutations already present, relative to the other WT that still accepts the given amino acid. The caveat being this is entrenchment based on a direct mutational path between the two homologs instead of some common ancestor, and to address this we have clarified our use of the term entrenchment on page 15.

4) “compensations of entrenched residues are not trivially pairwise … (Fig. 5e)”

Recent papers make it clear that mutations interact with many epistatic partners. (The authors cite such papers in the description of Fig. 5.) This means that a long-term change in the effect of a mutation is typically attributable to many substitutions, not a single or even a few. The authors’ finding is fully in line with this knowledge.

We agree with the reviewer’s statement. We have better highlighted that this is existing knowledge in the abstract, introduction (last paragraph), and discussion (second to last paragraph).

5) “the extent of heterogeneity in mutational effects uncovered by DCA and backed up by DMS is unprecedented”

We do not fully understand what “unprecedented” means, but here are two studies that show pervasive epistasis among distantly related homologs. PMID 29626131 compares two proteins that differ at 168 out of 365 sites (46% divergent), and finds that 63 of the 168 differences cause incompatibility. PMID 20975933 compares homologs that are up to 50% divergent and finds that the effects of a large set of mutations have become randomized. In terms of the extent of epistasis, it is not clear to us what is “unprecedented” about this study.

This may have been poor phrasing. “Unprecedented” refers to the degree to which such effects have been revealed by our method of combining DCA and DMS as independent and complementary data sources, which we note in the introduction is somewhat novel.

Overall, it is not that we doubt the value of this work. Far from it, we think the deployment of DCA and its conjunction with DMS is an exciting methodological direction. But it is the authors’ responsibility to fully develop the implications of their findings. In our opinion, the field of epistasis is inundated with studies that simply report the existence and prevalence of epistasis. It would be a service to the field for leaders such as the present authors to push beyond the mannerism and show what conceptual advances can be made.

We thank the reviewer for the positive note regarding our method for combining DCA and DMS, and indeed this is the key innovation we are trying to demonstrate in our study. We hope the revisions more clearly distinguish our contributions from existing knowledge.

REVIEWERS' COMMENTS

Reviewer #3 (Remarks to the Author):

We appreciate the authors' forthright response to our comments. The authors make an important stride in the study of protein evolution by harnessing the vast sequence information and computational capacity now available. We have no further concern to raise on the manuscript.